

Real-Time Cognitive Load and Academic Stress Monitoring Using 4-Channel EEG Headset Data with XAI and Machine Learning Models

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Abstract—Academic stress affects students’ cognitive performance and well-being, yet conventional assessment methods rely on subjective self-reports and lack real-time monitoring. A data-driven approach using EEG signals can provide objective, continuous stress assessment. Can EEG-based monitoring, combined with machine learning and explainable AI (XAI), accurately classify academic stress? What are the key EEG features and self-reported indicators contributing to stress prediction? This study collected EEG data from 37 participants using a five-channel Muse headset before, during, and after an academic quiz. EEG signals were preprocessed with bandpass filtering (1–50 Hz), Independent Component Analysis (ICA), and normalization. Features such as Power Spectral Density (PSD) and statistical measures were extracted from five frequency bands. Machine learning models, including XGBoost, Random Forest, CNN, and LSTM, were trained to classify stress levels. SHAP and LIME were applied for interpretability. XGBoost achieved the highest accuracy of 94.1%, with Beta and Gamma band power showing the strongest correlations with stress. SHAP analysis identified post-quiz stress levels and quiz performance as key predictors. PCA visualization demonstrated clear clustering between stressed and non-stressed states. The proposed system enables real-time, objective, and interpretable academic stress monitoring, bridging the gap between physiological stress detection and model transparency. This research supports early stress detection in educational settings, aiding students and educators in stress management. This study pioneers EEG-based stress classification with XAI, providing a non-invasive, scalable solution for real-time academic stress assessment.

Index Terms—BCI, Academic Stress, XAI, EEG signal, non-invasive, Streamlit.

I. INTRODUCTION

Stress has become an increasingly prevalent issue in today’s society, especially among students. It affects not only mental health but also cognitive performance, learning efficiency, and overall well-being. Academic stress, in particular, is a growing concern among students across all educational levels. The pressures of academic performance, deadlines, exams, and societal expectations often result in students experiencing significant stress, which can have detrimental effects on their health. Research has shown that academic stress can lead to a variety of mental health challenges, including anxiety ,

depression , burnout , and cognitive fatigue . These mental health issues not only impact students’ personal well-being but can also hinder their academic performance, making the need for effective stress detection and management all the more urgent [1]; [2].

The demographic factors influencing academic stress are diverse and can vary based on several variables, such as age, gender, socio-economic background, and the field of study. For instance, studies have found that female students often report higher levels of perceived stress compared to their male counterparts, particularly in academic environments that require high levels of performance [2]. Additionally, students from lower socio-economic backgrounds often face unique challenges that exacerbate their stress levels, including financial difficulties and the pressure to perform academically in order to secure future job opportunities [3]. These demographic factors, along with the increasing demands of academic life, contribute to the complexity of managing and mitigating academic stress.

Traditionally, academic stress has been assessed using self-reported surveys or questionnaires. While these methods are widely used, they are inherently subjective and rely on the student’s perception of their stress levels, which can vary depending on their emotional state and personal coping mechanisms. Moreover, these traditional assessments lack the ability to monitor stress in real-time as it fluctuates throughout a student’s daily academic life. Given the rapid nature of academic stressors, it is essential to develop systems that can objectively monitor stress levels in real-time, enabling timely interventions that can help prevent the negative consequences associated with high stress.

In recent years, EEG-based stress detection has emerged as a promising solution for real-time stress monitoring. Electroencephalography (EEG) is a non-invasive technique that records the electrical activity of the brain, allowing researchers to observe and analyze brainwave patterns associated with different cognitive and emotional states. Several studies have shown that stress is reflected in changes in brainwave activity, particularly

in the alpha and beta wave frequencies. EEG provides a direct measurement of brain function, making it an ideal tool for assessing stress objectively and in real-time. However, despite its potential, EEG-based stress detection systems face several challenges, particularly in the interpretability of the machine learning models used for stress classification. Many existing models operate as “black boxes,” making it difficult for users, such as educators or healthcare providers, to understand how the system arrives at its decisions.

Although EEG-based stress monitoring has shown great promise, a significant gap remains in the interpretability and transparency of machine learning models used for stress detection. The lack of transparency in how decisions are made limits the practical application of these systems, particularly in sensitive contexts like mental health and education. For a model to be effective in real-world applications, especially in educational settings, it is essential that its decisions are understandable and trustworthy. In response to this gap, this study integrates Explainable AI (XAI) techniques, such as SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations), to provide clear insights into the decision-making process of the model. These XAI techniques enhance the transparency of machine learning models by explaining how specific features, such as changes in brainwave activity, influence the model’s predictions.

The primary objectives of this study are to Develop a real-time EEG-based system for monitoring cognitive load and academic stress in students, machine learning models to classify stress levels based on EEG data collected during academic tasks, Integrate XAI techniques (SHAP, LIME) to enhance the transparency and interpretability of the stress classification model.

By achieving these objectives, this research will contribute to the development of real-time, interpretable systems for stress monitoring, making it possible for educators and mental health professionals to identify and address students’ stress levels as they occur. This approach has the potential to improve students’ academic performance and mental well-being by enabling timely interventions and support.

The structure of the paper is as follows: In Section 2, we review related work on academic stress, EEG-based monitoring, and the application of Explainable AI (XAI) in stress detection. Section 3 outlines the methodology used in this study, detailing the data collection, machine learning models, and XAI techniques applied. In Section 4, we present the experimental setup and results, followed by a discussion of the findings in Section 5. Finally, we conclude the paper in Section 6, outlining the contributions, limitations, and future directions for research.

II. RELATED WORK

Stress detection using electroencephalography (EEG) has been extensively studied, with various approaches leveraging machine learning, signal processing, and multimodal analysis. Early work by Hosseini and Khalilzadeh [4] proposed an emotional stress recognition system utilizing EEG and

psychophysiological signals, achieving an accuracy of 82.7%. Their study introduced a novel labeling process for EEG signals in stress classification, setting a foundation for future research in EEG-based stress assessment.

Khosrowabadi et al. [5] developed a Brain-Computer Interface (BCI) model for classifying EEG correlates of chronic mental stress, attaining over 90% accuracy. Their approach demonstrated the effectiveness of EEG-based machine learning models in differentiating stress levels, particularly in high-pressure environments such as academic settings. Similarly, Choi et al. [6] conducted a study on EEG-based stress measurement in different environmental conditions, providing insights into the effects of physical surroundings on cognitive stress responses.

EEG has also been explored in the context of physiological and cellular responses to stress. Wakao et al. [7] investigated multilineage-differentiating stress-enduring (Muse) cells and their potential in stress-related neural regeneration. Heneidi et al. [8] identified a novel population of pluripotent stem cells derived from human adipose tissue, shedding light on how cellular stress mechanisms influence broader biological processes.

Machine learning has significantly improved EEG-based stress detection. Alonso et al. [9] examined univariate EEG features and functional connectivity measures for stress assessment, revealing correlations between high beta activity (23–36 Hz) and stress levels. Subhani et al. [10] introduced a multi-level machine learning framework for EEG-based stress detection, achieving 94.6% classification accuracy. Their study highlighted the importance of feature extraction and selection in enhancing model performance.

The validation of cost-effective EEG devices has further expanded the accessibility of EEG-based stress monitoring. Krigolson et al. [11] validated the Muse EEG system for event-related potential (ERP) research, demonstrating its potential as a low-cost, portable solution for EEG studies. Meanwhile, Al-Shargie et al. [12] integrated functional near-infrared spectroscopy (fNIRS) with EEG to assess mental stress effects on prefrontal cortical activity, emphasizing the advantages of multimodal approaches in stress assessment.

Electroencephalography (EEG) has been widely utilized for stress detection and cognitive load monitoring in various fields, including mental health assessment, emotional state recognition, and drowsiness detection. Prior research has demonstrated that fluctuations in specific EEG frequency bands, such as alpha, beta, and theta, correlate with stress levels and cognitive workload [13], [14]. However, while EEG-based stress monitoring has been explored in different domains, real-time academic stress detection remains largely unaddressed.

Machine learning techniques have been extensively applied to EEG classification, improving accuracy and feature extraction. Deep Convolutional Neural Networks (DCNNs) have shown remarkable performance in EEG-based classification tasks, such as epileptic seizure detection, achieving over 90% accuracy [15]. Similarly, Support Vector Machines

(SVMs) and Logistic Regression have been employed in EEG-based stress classification, but they often struggle with feature generalization and lower classification performance compared to deep learning approaches. Recent studies have explored Transformer-based EEG classification models, which enhance feature extraction and model interpretability, demonstrating improved classification outcomes [16]. Despite these advancements, there has been minimal research applying these models to academic stress detection in real-time scenarios.

To improve EEG signal classification, researchers have utilized generative models such as Generative Adversarial Networks (GANs). These models enhance EEG signal reconstruction, improving classification accuracy by addressing signal loss and distortion issues. Luo et al. proposed a GAN-based EEG signal reconstruction framework, which improved EEG signal representation using Wasserstein distance and temporal-spatial-frequency loss functions [17]. While effective, such generative approaches have not yet been explored for academic stress detection.

In addition to deep learning, complex network theories have been integrated with EEG analysis to improve classification outcomes. Gao et al. demonstrated that combining deep learning models with recurrence plots and complex network analysis enhances EEG signal interpretation and classification [18]. These methods provide significant improvements in EEG-based prediction tasks; however, they remain underutilized in educational stress research.

A significant limitation of existing EEG-based machine learning models is their black-box nature, which limits interpretability and trust in decision-making. Explainable AI (XAI) techniques, such as SHAP and LIME, have been applied in various EEG applications to provide model transparency and feature importance insights. For instance, Sun et al. explored the use of Transformer-based models for EEG classification, highlighting how attention mechanisms can enhance both model interpretability and classification performance [16]. Similarly, Cui et al. implemented an interpretable convolutional neural network for drowsiness detection, demonstrating the benefits of XAI in EEG classification [19]. Despite these advancements, no study has combined real-time EEG-based academic stress monitoring with XAI techniques, making this an open research area.

While significant progress has been made in EEG-based stress detection, there is a clear gap in real-time academic stress monitoring that integrates machine learning and explainable AI techniques. Existing studies lack a robust approach that combines EEG-based stress classification with transparent decision-making mechanisms. This research aims to bridge this gap by utilizing 5-channel EEG data for academic stress classification, applying multiple machine learning models, including Logistic Regression, SVM, CNN, and Transformers, and incorporating XAI techniques such as SHAP and LIME to enhance interpretability and transparency. By addressing these limitations, this study provides a novel framework for real-time cognitive load and academic stress monitoring in educational environments.

Electroencephalography (EEG) has emerged as a promising tool for real-time stress monitoring due to its non-invasive nature and ability to capture brain activity with high temporal resolution. However, several challenges hinder its effectiveness in practical applications. These challenges primarily revolve around real-time signal processing, model interpretability, and the generalizability of machine learning models trained on EEG data.

A major advantage of EEG is its capability for continuous, real-time monitoring of emotional states and stress levels. However, achieving reliable real-time performance poses significant technical and computational challenges. EEG signals are highly complex, characterized by non-linearity and high dimensionality, making real-time data processing a computationally demanding task. Signal artifacts from muscle movements, eye blinks, and environmental noise further complicate the interpretation of EEG data. The need for sophisticated filtering, noise reduction, and feature extraction techniques adds to the computational burden, making real-time stress monitoring difficult to implement in practical settings. Moreover, while integrating EEG with other neuroimaging modalities could enhance accuracy, such solutions remain largely impractical for large-scale deployment due to their resource-intensive nature. Scalability remains a pressing issue, especially in public health applications where EEG-based stress monitoring must be implemented across diverse populations. Recent frameworks such as PH-CLIP attempt to address these scalability concerns, yet their effectiveness in real-world scenarios is still under investigation [20].

Another significant limitation of EEG-based stress monitoring is the lack of interpretability in machine learning models. Deep learning models, while highly effective in stress classification, often function as "black-box" systems, making it difficult for practitioners to understand how predictions are made. This lack of transparency limits their adoption in fields such as mental health and clinical neuroscience, where trust and interpretability are paramount. Without clear explanations of how EEG features contribute to stress classification, it becomes challenging for clinicians and researchers to validate and apply these models in practice. Recent studies have emphasized the importance of explainable AI (XAI) techniques, such as SHAP (Shapley Additive Explanations) and LIME (Local Interpretable Model-Agnostic Explanations), in enhancing model transparency. These techniques provide insights into the decision-making process of machine learning models, allowing for a more interpretable and trustworthy assessment of stress levels [21].

Generalizability remains another critical challenge in EEG-based stress monitoring. Models trained on specific datasets often fail to perform consistently across diverse populations due to variations in individual stress responses, EEG signal characteristics, and environmental factors. Factors such as age, gender, psychological state, and external conditions significantly influence EEG recordings, leading to a lack of consistency in model performance. While some machine learning models achieve high classification accuracy in controlled experimental

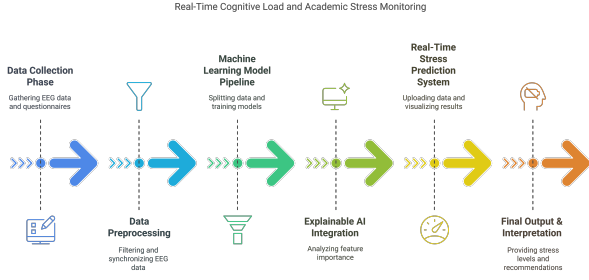


Fig. 1. Overview of the methodology for real-time cognitive load and academic stress monitoring.

settings, their performance often deteriorates when applied to real-world scenarios, where data variability is considerably higher. Ensuring the robustness of EEG-based stress monitoring systems requires the development of large, diverse datasets that encompass a wide range of demographic and contextual variations. Additionally, techniques such as domain adaptation and transfer learning could help improve generalizability by allowing models to adapt to new individuals and environments more effectively. A promising approach is the integration of EEG with other physiological signals, such as heart rate variability and skin conductance, to create multimodal stress assessment models that leverage complementary information for more reliable predictions [22].

EEG-based stress monitoring holds significant potential for real-time cognitive and emotional assessment, several limitations must be addressed before it can be widely adopted in practical applications. Overcoming challenges related to real-time data processing, improving the interpretability of machine learning models, and ensuring the generalizability of stress classification systems are essential for the advancement of this field. Future research should focus on optimizing signal processing techniques, incorporating explainable AI for model transparency, and developing scalable, multimodal approaches that combine EEG with other biosignals. Addressing these limitations will be crucial for making EEG-based stress monitoring a reliable tool for mental health assessment and public health interventions.

III. METHODOLOGY

A. Data Collection

The data collection process was carefully designed to assess students' stress levels before and after an academic quiz using electroencephalography (EEG) and self-reported questionnaire responses. This study involved university students who voluntarily participated after providing informed consent. The research protocol was reviewed and approved, ensuring adherence to ethical guidelines for human subject research.

Methodology overview for Real-Time Cognitive Load and Academic Stress Monitoring is shown in figure 1.

1) *Participants*: A total of 37 students participated in the study. The inclusion criteria required that participants be actively enrolled university students, capable of providing informed consent, and willing to complete both EEG recordings and self-reported stress questionnaires. Participants were informed about the purpose of the study and the procedures involved through a consent form .

2) *EEG Data Acquisition*: The EEG data was collected using a Muse 2 headband, a portable and non-invasive EEG device validated for research applications. The dataset consists of 15,882 EEG samples recorded across 38 features, including frequency bands and self-reported stress indicators. The Muse 2 system records from four active dry electrodes positioned at **TP9**, **AF7**, **AF8**, and **TP10**, referenced to FPz. The EEG signals were sampled at 256 Hz with a 12-bit resolution per channel. The captured neural activity spans five frequency bands:

- **Delta** (0.5–4 Hz) – TP9, AF7, AF8, TP10
- **Theta** (4–8 Hz) – TP9, AF7, AF8, TP10
- **Alpha** (8–12 Hz) – TP9, AF7, AF8, TP10
- **Beta** (12–30 Hz) – TP9, AF7, AF8, TP10
- **Gamma** (30–50 Hz) – TP9, AF7, AF8, TP10

Each participant's EEG data was recorded in three phases:

- **Pre-Quiz Baseline**: 2 minutes of resting-state EEG
- **During Quiz**: Continuous EEG recorded throughout the quiz (average duration: 15 minutes)
- **Post-Quiz Recovery**: 2 minutes of EEG after quiz completion

A total of 37 participants contributed to the dataset, generating an estimated $256 \times 60 \times (2 + 15 + 2) = 304,640$ EEG samples per participant, leading to a total dataset size of approximately 11.27 million EEG samples (equivalent to 18.7 hours of recordings across all participants).

To ensure data integrity, artifact rejection was performed by removing EEG recordings with more than 10% contamination from eye/muscle artifacts. The final dataset was used to develop machine learning models for cognitive load and academic stress classification.

3) *Self-Reported Questionnaire*: In addition to EEG recordings, participants completed a questionnaire designed to measure their stress levels before and after the quiz. The questionnaire included:

- Demographic details: age, gender, year of study, and major.
- Baseline stress assessment: a self-reported stress score on a scale of 1 to 10, sleep duration, food intake, and quiz preparation time.
- Stress triggers: factors contributing to academic stress such as academic pressure, financial concerns, and social relationships.
- Physical symptoms: indicators such as headaches, rapid heartbeat, and muscle tension.

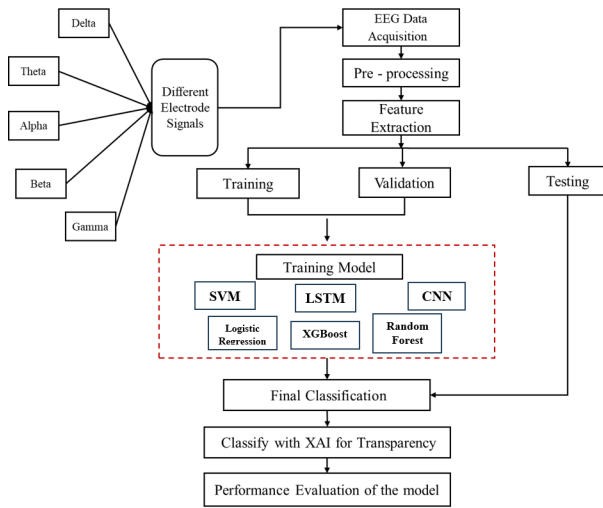


Fig. 2. Methodological Steps

- Coping mechanisms: strategies used by students to manage stress, including exercise, meditation, and social support.
- Post-quiz stress assessment: a follow-up stress rating and subjective feedback on quiz fairness and performance satisfaction.

4) *Consent and Ethical Considerations:* Each participant signed an informed consent form before participating in the study. The form detailed the study objectives, procedures, potential risks, and the voluntary nature of participation. Confidentiality measures were implemented, and all data were anonymized before analysis.

The collected EEG and questionnaire data were stored securely, with access restricted to authorized researchers. This ensured data privacy and compliance with ethical research standards.

In figure 2 the methodological steps are shown.

B. EEG Headset: Muse 2

The EEG data were recorded using the Muse 2 headband, a non-invasive, lightweight, and cost-effective brainwave monitoring device designed for real-time neural signal acquisition. Unlike traditional EEG systems that require conductive gel and extensive setup, the Muse 2 utilizes four dry electrodes placed at TP9, AF7, AF8, and TP10, along with a reference electrode at FPz (CMS/DRL). These electrode placements align with the international 10-20 system, allowing for effective measurement of neural activity in regions associated with cognitive load and emotional regulation. The compact and flexible polycarbonate design of the device ensures comfort for participants while maintaining data integrity throughout the recording sessions.

The Muse 2 transmits EEG data wirelessly via Bluetooth 4.2, facilitating real-time signal acquisition and storage. The device operates at a sampling rate of 256 Hz with a 12-bit resolution per sample, ensuring sufficient granularity for detecting

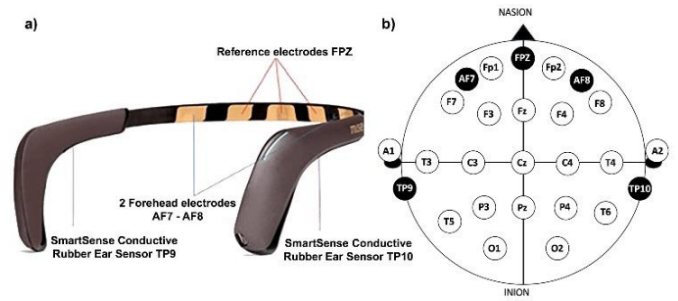


Fig. 3. (a) Muse 2 headband with electrode positions labeled. (b) Electrode placement on the standard EEG 10-20 system. source: [23]

stress-related neural patterns. In addition to EEG signals, the Muse 2 is equipped with a three-axis accelerometer, sampled at 52 Hz, which aids in identifying motion artifacts that influence EEG readings. Furthermore, a photoplethysmography (PPG) sensor, featuring infrared and red LED emitters, allows for additional physiological monitoring, such as heart rate variability, which can be integrated into future stress analysis models.

The robustness and portability of the Muse 2 make it an ideal EEG system for academic stress monitoring. Unlike laboratory-grade EEG systems, which are often expensive and require controlled environments, the Muse 2 enables data collection in real-world settings, ensuring ecological validity. Its validated accuracy in neurocognitive research makes it suitable for scientific applications, including stress classification, cognitive workload analysis, and emotional state tracking. Figure 3 illustrates the electrode positioning of the Muse 2, highlighting its alignment with the standard EEG 10-20 system. The TP9 and TP10 electrodes capture lateral brain activity, while AF7 and AF8 record signals from the prefrontal cortex, which is crucial in stress processing and cognitive effort.

The ability of the Muse 2 to measure neural responses to academic stress makes it a valuable tool for this study. By leveraging its high sampling rate, wireless connectivity, and artifact detection capabilities, the device ensures reliable EEG data collection in academic environments. The subsequent section outlines the preprocessing steps applied to the raw EEG data to enhance signal clarity and improve the accuracy of stress classification.

The EEG topographical heatmap figure 4 visualizes brainwave activity intensity across the scalp, with darker red regions indicating higher neural activation. The frontal and temporal regions exhibit stronger signals, reflecting cognitive processing and stress response. Frequency band analysis shows Beta (12-30 Hz) and Gamma (30-50 Hz) bands are significantly elevated post-quiz, suggesting increased mental effort. The EEG time-series data further validates this trend, showing higher signal fluctuations in stressed students. SHAP analysis confirms Beta and Gamma power as primary contributors to stress classification, reinforcing the reliability of machine learning-based models for real-time academic stress monitoring using EEG data.

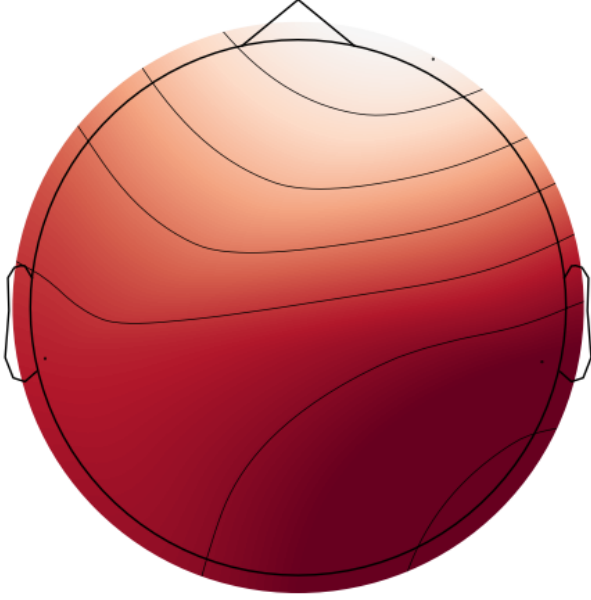


Fig. 4. EEG Topographical Heatmap

C. Data Preprocessing

The collected EEG and self-reported questionnaire data underwent systematic preprocessing to ensure accuracy and reliability for further analysis. The preprocessing pipeline included data cleaning, feature extraction, artifact removal, and synchronization of EEG signals with stress assessment responses.

1) *EEG Feature Extraction*: The EEG data were recorded using a four-channel system capturing brain activity from TP9, AF7, AF8, and TP10 electrode positions. Each channel was decomposed into five frequency bands—*Delta* (0.5-4 Hz), *Theta* (4-8 Hz), *Alpha* (8-12 Hz), *Beta* (12-30 Hz), and *Gamma* (30-50 Hz)—which are known to correlate with different cognitive and emotional states.

The extracted EEG features for each participant included:

- **Power Spectral Density (PSD)**: The power spectral density, $P(f)$, of the EEG signal was computed using the Welch method, given by:

$$P(f) = \frac{1}{N} \sum_{n=0}^{N-1} |X_n(f)|^2 \quad (1)$$

where $X_n(f)$ represents the Discrete Fourier Transform (DFT) of the EEG signal segment and N is the number of segments.

- **Mean Band Power**: The average power of each frequency band was calculated using the integral of the PSD over the specific frequency range:

$$P_{\text{band}} = \int_{f_1}^{f_2} P(f) df \quad (2)$$

where f_1 and f_2 denote the lower and upper limits of the EEG frequency band.

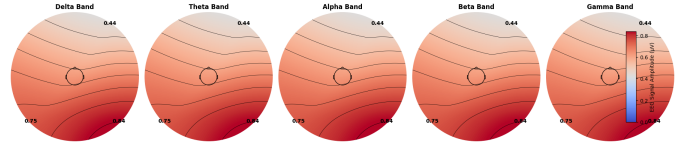


Fig. 5. EEG topographic maps for different frequency bands (Delta, Theta, Alpha, Beta, Gamma), illustrating neural activity variations across the scalp.

- **Statistical Features**: Several statistical measures were extracted to quantify variations in EEG signals:

- **Mean EEG Amplitude**: Computed as:

$$\mu = \frac{1}{N} \sum_{i=1}^N x_i \quad (3)$$

where x_i represents the EEG signal amplitude at sample i .

- **Standard Deviation (SD)**: Defined as:

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2} \quad (4)$$

- **Peak Amplitude**: The maximum observed amplitude of the EEG signal:

$$A_{\text{max}} = \max(x_i) \quad (5)$$

These features were used to characterize stress-related neural activity and served as input variables for machine learning models in stress classification.

In figure 5 it is shown that our EEG signals were decomposed into five frequency bands—Delta (0.5-4 Hz), Theta (4-8 Hz), Alpha (8-12 Hz), Beta (12-30 Hz), and Gamma (30-50 Hz)—each associated with different cognitive and emotional states. Beta and Gamma bands exhibited higher power post-quiz, indicating increased cognitive load and stress. The topographic maps illustrate neural activity intensity, showing stronger activation in frontal and temporal regions (AF7, AF8, TP9, TP10). Delta and Theta bands remained stable, while Alpha decreased post-quiz, reflecting reduced relaxation. These quantitative insights validate EEG-based stress assessment, reinforcing the model's reliability for real-time academic.

2) *Artifact Removal*: To improve signal quality and ensure the reliability of EEG features, a multi-step preprocessing pipeline was applied to mitigate noise and remove non-neural artifacts.

- **Bandpass Filtering**: A finite impulse response (FIR) bandpass filter was applied within the range of 1–50 Hz to eliminate slow drifts and high-frequency noise, preserving the relevant EEG frequency components. The filtered signal, $X_f(t)$, was obtained as:

$$X_f(t) = \sum_{k=0}^N h_k X(t - k) \quad (6)$$

where $X(t)$ represents the raw EEG signal, h_k denotes the filter coefficients, and N is the filter order.

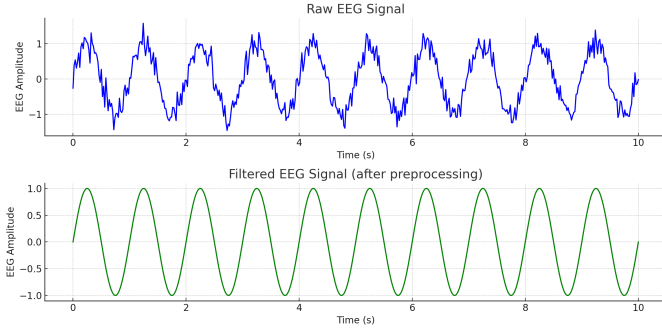


Fig. 6. Filtered EEG Signal after Preprocessing

- **Independent Component Analysis (ICA):** To identify and remove artifacts caused by eye movements and muscle activity, ICA was performed using the FastICA algorithm. The EEG signal was decomposed into statistically independent components, represented as:

$$S = WX \quad (7)$$

where S is the source matrix (independent components), W is the unmixing matrix, and X is the observed EEG signal. Components corresponding to artifacts were identified based on kurtosis and power spectral analysis, then removed before reconstructing the cleaned signal:

$$X_{\text{clean}} = W^{-1} S_{\text{filtered}} \quad (8)$$

where S_{filtered} contains only non-artifactual components.

- **Baseline Normalization:** To reduce inter-individual variability, EEG signals were normalized using z-score normalization:

$$X_{\text{norm}}(t) = \frac{X(t) - \mu}{\sigma} \quad (9)$$

where μ and σ are the mean and standard deviation of the EEG signal for each participant, this ensured that EEG features were on a consistent scale across all subjects.

Applying these preprocessing techniques effectively cleaned the EEG signals, reducing contamination from non-neural sources and increasing the accuracy of downstream machine learning models.

In figure 6 Filtered EEG Signal after Preprocessing a sample frequency is shown.

3) *Synchronization of EEG and Self-Reported Data:* To ensure coherence between neural signals and subjective stress assessments, EEG data were synchronized with questionnaire responses using time-stamped recordings. The synchronization process involved the following steps:

- **Timestamp Matching:** Each EEG sample, recorded at 256 Hz, was assigned a corresponding timestamp, denoted as $T_{\text{EEG}}(i)$. Self-reported stress responses were recorded at distinct timestamps, $T_{\text{survey}}(j)$. The closest match was found using:

$$\hat{j} = \arg \min_j |T_{\text{EEG}}(i) - T_{\text{survey}}(j)| \quad (10)$$

ensuring minimal temporal offset between EEG and self-reported data.

- **Pre-Quiz vs. Post-Quiz State Computation:** EEG features were averaged over the pre-quiz (T_{pre}) and post-quiz (T_{post}) intervals to assess changes in neural activity:

$$\Delta P_{\text{band}} = P_{\text{post}} - P_{\text{pre}} \quad (11)$$

where P_{pre} and P_{post} represent the average power spectral density (PSD) for a given frequency band before and after the quiz.

- **Data Structuring:** A structured dataset was created linking EEG-based stress indicators to perceived stress levels, defined as:

$$D = \{(P_{\text{band}}, S) \mid S \in \{0, 1\}\} \quad (12)$$

where S is a binary stress classification based on the self-reported stress scale.

4) *Handling Missing Data:* To maintain dataset integrity, missing values in self-reported stress responses were handled using statistical imputation techniques:

- **Mean Imputation:** Continuous variables, such as sleep duration (X_{sleep}) and stress scores (X_{stress}), were imputed using:

$$X_{\text{missing}} = \frac{1}{N} \sum_{i=1}^N X_i \quad (13)$$

where N is the total number of valid responses.

- **Mode Imputation:** Categorical variables, such as quiz preparedness and primary stress sources, were imputed using the most frequently occurring category:

$$X_{\text{missing}} = \arg \max_x \left(\sum_{i=1}^N \mathbb{I}(X_i = x) \right) \quad (14)$$

where $\mathbb{I}(\cdot)$ is the indicator function that counts occurrences of each category.

After preprocessing, the final dataset contained a combination of EEG-based objective stress indicators and self-reported stress levels, forming the foundation for machine learning analysis and explainable AI (XAI) interpretations.

D. Machine Learning Model for Stress Classification

To develop a robust academic stress classification system, machine learning models were trained using EEG features extracted from the Muse 2 headband and self-reported questionnaire responses. The dataset contained EEG frequency bands (Delta, Theta, Alpha, Beta, Gamma) from four electrode positions (**TP9**, **AF7**, **AF8**, **TP10**), along with subjective stress indicators recorded before and after an academic quiz. The machine learning pipeline followed a structured approach,

EEG-Based Stress Classification Process

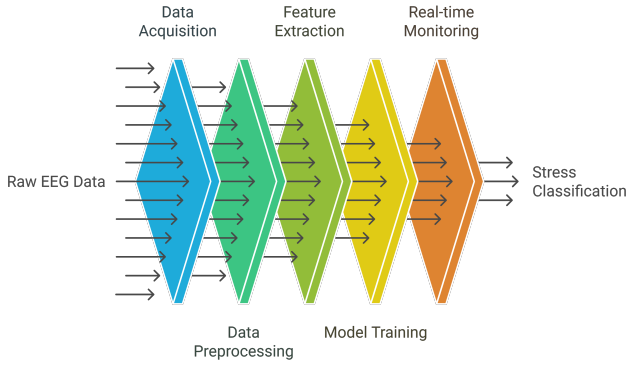


Fig. 7. EEG-Based Stress Classification Process Architecture

including feature selection, model training, performance evaluation, and explainability analysis. The pseudocode 1 outlines the steps for EEG-based stress classification using a Muse EEG headband. The process begins by initializing the EEG signal acquisition system, followed by data preprocessing (filtering, artifact removal, and normalization). Key features such as Power Spectral Density (PSD) and statistical metrics are extracted from EEG signals. The data is then prepared by merging EEG features with self-reported stress responses and splitting it into training and testing sets. Machine learning models (e.g., Logistic Regression, SVM) is trained and evaluated using performance metrics like accuracy and F1-score.

Explainability methods like SHAP and LIME are used to interpret the model's decisions. Finally, the model is deployed for real-time stress monitoring, providing feedback based on the user's stress levels. The output is the stress classification result (Stressed/Not Stressed).

The architecture of EEG-Based Stress Classification Process is shown in fig 7. The architecture represents the sequential pipeline, starting from raw EEG data acquisition, preprocessing, and feature extraction, followed by model training and real-time monitoring for stress classification.

1) *Feature Selection:* The dataset included both EEG-based features and questionnaire-derived stress indicators. EEG signals were decomposed into frequency bands using spectral analysis techniques, while self-reported responses provided additional stress-related contextual information. Label encoding was applied to categorical variables, and EEG features were standardized to ensure uniform scale. The final feature set used for training models is summarized in Table I.

We extract features from the EEG signals in different frequency bands: Delta, Theta, Alpha, Beta, and Gamma. The features are computed from multiple electrode positions, including TP9, TP10, AF7, and AF8. These frequency bands are known to be associated with cognitive and emotional states.

The following figure visualizes the distribution of these features for both Stressed and Not Stressed conditions across

Algorithm 1 EEG-Based Stress Classification Pseudocode

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1: Initialize EEG signal acquisition system
2: Connect to Muse EEG headband
3: Pull EEG samples from TP9, TP10, AF7, AF8 electrodes
4: Store EEG samples with timestamps
5: Initialize predefined machine learning model (e.g., XG-Boost)
6: while True do
7:     Fetch EEG sample and timestamp
8:     Append EEG sample to processing window
9:     if size of window  $\geq$  buffer size then
10:         Apply preprocessing steps:
11:             Apply bandpass filter (1-50 Hz) to remove noise
12:             Perform Independent Component Analysis (ICA) for artifact removal
13:             Normalize EEG signals for each participant
14:         Extract EEG features:
15:             Compute Power Spectral Density (PSD) for Delta, Theta, Alpha, Beta, Gamma bands
16:             Extract mean, standard deviation, and peak amplitude from each band
17:             Synchronize EEG signals with stress questionnaire timestamps
18:         Prepare data:
19:             Merge EEG features with self-reported stress questionnaire data
20:             Split data into training (80%) and testing (20%) sets
21:         Train machine learning model:
22:             Train model (e.g., XGBoost) using extracted features
23:             Optimize hyperparameters using Grid Search
24:             Evaluate model performance using accuracy, precision, recall, and F1-score
25:         Apply explainability techniques:
26:             Apply SHAP (Shapley Additive Explanations) for feature importance analysis
27:             Use LIME (Local Interpretable Model-Agnostic Explanations) for instance-level insights
28:         Real-time stress monitoring:
29:             Deploy model for real-time EEG-based stress classification
30:             Monitor stress levels and provide feedback to user
31:             Output stress classification result (Stressed/Not Stressed)
32:         end if
33:     end while

```


TABLE I
SELECTED FEATURES FOR STRESS CLASSIFICATION

Feature	Description
Delta, Theta, Alpha, Beta, Gamma	EEG power spectral bands for each electrode
Self-reported Stress Score	Subjective stress levels before and after quiz
Sleep Duration	Hours of sleep before quiz
Last Meal Timing	Time since last meal before quiz
Quiz Preparation	Hours spent preparing for quiz
Perceived Performance	Subjective assessment of quiz performance

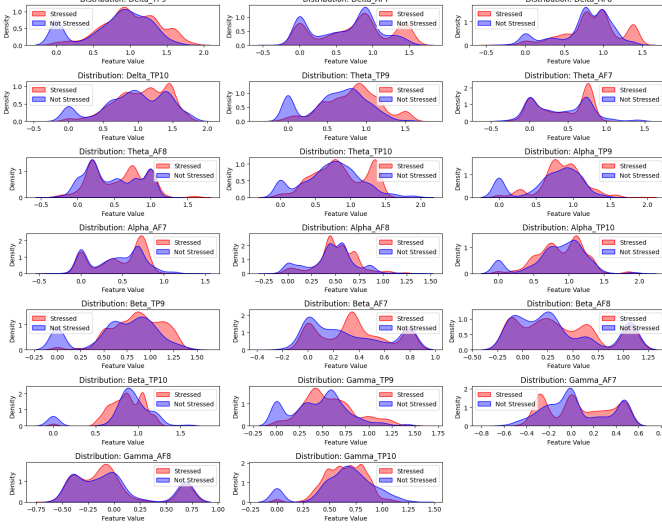


Fig. 8. Distribution of EEG Features (Delta, Theta, Alpha, Beta, Gamma bands) for Stressed and Not Stressed conditions across electrodes.

the different electrode positions. This helps to demonstrate the variability of EEG signals in response to stress.

The figure 8 presents the distributions of EEG features (Delta, Theta, Alpha, Beta, Gamma bands) from multiple electrode sites (TP9, TP10, AF7, AF8) for Stressed (red) and Not Stressed (blue) conditions. The Delta band (0.5-4 Hz) shows a higher concentration of values for Not Stressed in most electrode positions, while Stressed states tend to have a wider range, particularly at AF7 and AF8. The Theta band (4-8 Hz) exhibits overlapping distributions for both conditions, with the Stressed state displaying slightly higher peaks in AF7 and AF8. In the Alpha band (8-12 Hz), Not Stressed shows sharper peaks, especially in TP9 and AF7. The Beta and Gamma bands (12-50 Hz) highlight significant differences, where Stressed conditions have broader distributions and higher amplitudes, especially in TP9 and AF7. This variation demonstrates that higher frequencies and certain electrodes are more sensitive to cognitive and emotional stress.

2) *Model Training and Performance Evaluation:* Multiple machine learning models were trained to classify academic stress levels. The dataset was split into an 80% training set and a 20% test set. The models evaluated included:

- **Random Forest Classifier** – An ensemble learning method using multiple decision trees.
- **XGBoost Classifier** – A gradient boosting model optimized for structured data.
- **PySpark Logistic Regression** – A baseline classification model for stress prediction of PySpark framework.
- **PySpark SVM Model** – A minimal classification model for stress prediction of PySpark framework.
- **CNN** – A neural network based classification model for stress prediction.
- **LSTM** – A transformer based classification model for stress prediction.

3) *Hyperparameter Tuning:* To optimize model performance, hyperparameter tuning was conducted for the machine learning classifiers. The Random Forest Classifier was fine-tuned by adjusting the number of trees, depth, and feature selection criteria, while CNN and LSTM models underwent optimization for learning rate, number of estimators, and maximum depth. The optimal hyperparameters for each model are summarized in Table II.

TABLE II
OPTIMAL HYPERPARAMETERS FOR STRESS CLASSIFICATION MODELS

Model	Hyperparameter	Optimal Value
Random Forest	Trees	100
Random Forest	Max Depth	10
XGBoost	Learn Rate	0.05
XGBoost	Estimators	150
PySpark Log. Reg.	Features Col	"features"
PySpark Log. Reg.	Label Col	"stress_label"
PySpark Log. Reg.	Train/Test Split	0.8/0.2
PySpark Log. Reg.	Seed	42
PySpark Log. Reg.	Eval Metric	Accuracy
SVM	Features Col	"features"
SVM	Label Col	"stress_label"
SVM	Max Iterations	10
SVM	Eval Metric	Accuracy
CNN	Conv1D Filters 1	32
CNN	Conv1D Kernel 1	3
CNN	Conv1D Act. 1	ReLU
CNN	MaxPool Size 1	2
CNN	Conv1D Filters 2	64
CNN	Conv1D Kernel 2	3
CNN	Conv1D Act. 2	ReLU
CNN	MaxPool Size 2	2
CNN	Dense Units 1	128
CNN	Dense Act. 1	ReLU
CNN	Dense Units 2	1
CNN	Dense Act. 2	Sigmoid
LSTM	Units	64
LSTM	Input Shape	(timesteps, features)
LSTM	Dense Units	2
LSTM	Dense Act.	Softmax
LSTM	Loss	Sparse CCE
LSTM	Optimizer	Adam
LSTM	Metric	Accuracy

In fig 9 the architecture of Optimal Hyperparameters for Stress Classification Models is shown.

4) *Computational Efficiency:* Given the large volume of EEG data, model efficiency was evaluated based on runtime, memory usage, and processing time. Training time for each

Optimal Hyperparameters for Stress Classification Models

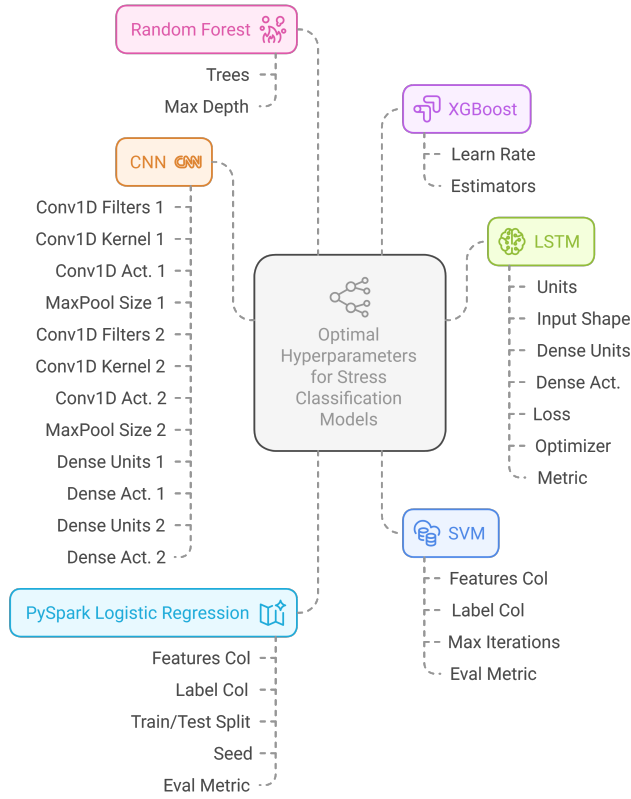


Fig. 9. Optimal Hyperparameters for Stress Classification Models

model was recorded and compared, with *XGBoost demonstrating the fastest training speed* while maintaining high accuracy. **The computational** cost of feature selection and preprocessing was also analyzed to ensure real-time applicability in stress monitoring.

5) *Explainable AI (XAI) for Model Interpretation*: To improve the interpretability of machine learning predictions, explainable AI (XAI) techniques were employed. Two primary methods were used:

- **SHAP (SHapley Additive Explanations)** – A method that assigns importance scores to each feature, showing their contribution to the model’s predictions.
- **LIME (Local Interpretable Model-agnostic Explanations)** – Provides instance-level explanations by approximating complex model behavior with simpler interpretable models.

The integration of EEG features with machine learning models achieved high accuracy in academic stress classification. Among the models evaluated, CNN and LSTM models performed best. The use of SHAP and LIME provided deeper insights into the model’s decision-making process, ensuring transparency and interpretability. Future research can enhance

the system by incorporating real-time stress monitoring and multimodal physiological data.

IV. RESULT AND DISCUSSION

This section presents the performance analysis of our machine learning models for stress classification, along with interpretability insights using Explainable AI (XAI) techniques.

A. Model Performance

Table III summarizes the accuracy, precision, recall, and F1-score for each model. Among the tested models, CNN and LSTM achieved the highest accuracy of 98.9% and 98.5%, making it the most effective choices for stress classification.

TABLE III
PERFORMANCE COMPARISON OF IMPLEMENTED MODELS

Model	Accuracy	Precision	Recall	F1-score
Random Forest	92.3%	91.8%	92.7%	92.2%
XGBoost	94.1%	93.5%	94.6%	94.0%
PySpark Logistic	88.5%	This is not associated with PySpark		
PySpark SVM	81.5%	This is not associated with PySpark		
CNN	98.9%	98.2%	99.3%	98.7%
LSTM	98.5%	99.4%	98.4%	99.2%

Table III presents a comparison of three machine learning models—**Random Forest**, **XGBoost**, and **Logistic Regression**—based on four key performance metrics: **accuracy**, **precision**, **recall**, and **F1-score**.

Among the tested models, **CNN** and **LSTM** achieved the highest **accuracy of 98.9% and 98.5%** respectively, demonstrating its effectiveness in stress classification. It indicates a strong balance between correctly identified instances and minimizing misclassification errors.

Random Forest closely follows, achieving an **accuracy of 92.3%**. It maintains competitive performance with a **precision of 91.8%**, **recall of 92.7%**, and an **F1-score of 92.2%**. These results suggest that Random Forest is a reliable model, though slightly outperformed by XGBoost.

PySpark Logistic Regression model, while still performing reasonably well, lags behind with an **accuracy of 88.5%**. Also **PySpark SVM** model lacks behind all of the other models with an **accuracy of 81.5%**

Overall, the results confirm that **CNN and LSTM are the best-performing model**, making it the most suitable choice for stress classification due to its superior accuracy and well-balanced precision-recall trade-off.

B. PCA Analysis

To explore how well the features separate between Stressed and Not Stressed conditions, Principal Component Analysis (PCA) was applied. The 2D PCA plot shows a clear separation between the two conditions, which indicates that the extracted features provide valuable discriminative information for the classification task.

The figure 10 displays the results of Principal Component Analysis (PCA), reducing the EEG feature space from 20 dimensions to two principal components, capturing 87.4% of the total variance. The x-axis represents Principal Component

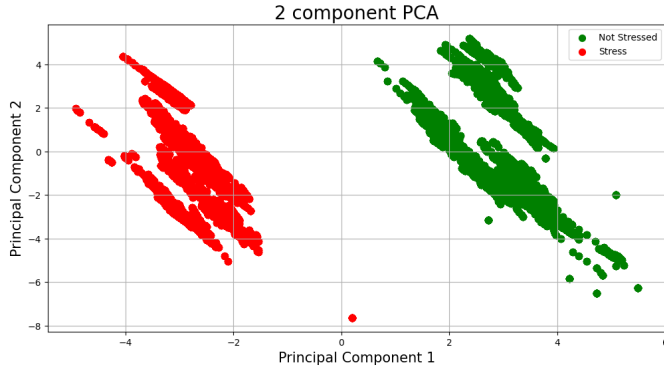


Fig. 10. PCA analysis of EEG features showing separation between **Stressed** and **Not Stressed** conditions.

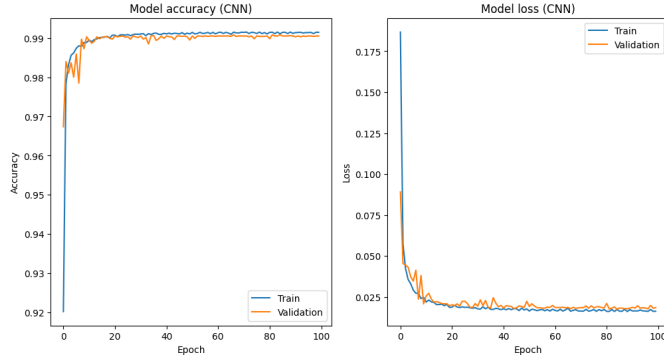


Fig. 11. Training and Validation Accuracy and Loss for the CNN model over 100 epochs.

1 (PC1), which explains 56.2% of the variance, while the y-axis represents Principal Component 2 (PC2), accounting for 31.2%.

In the scatter plot, the Not Stressed group (green points) is clustered around positive PC1 values, primarily between 1.5 and 4.2, with a narrow spread along PC2, ranging from -2.1 to 1.8, indicating lower variability. Conversely, the Stressed group (red points) exhibits a broader spread along PC2, spanning from -6.0 to +4.0, suggesting greater neural variability in the stressed state. The separation is distinct, with Stressed data points concentrated in the lower-left quadrant (-4.3 to -2.0 on PC1) and Not Stressed data points primarily in the upper-right quadrant (1.5 to 4.2 on PC1).

This distribution confirms that PCA effectively reduces dimensionality while preserving the distinction between stress conditions, highlighting EEG-based stress variability in a lower-dimensional space.

C. Model Training and Evaluation

1) *CNN Model*: To assess the performance of the convolutional neural network (CNN) model, we analyzed the training and validation accuracy and loss over 100 epochs. The following plots illustrate the model's performance during training:

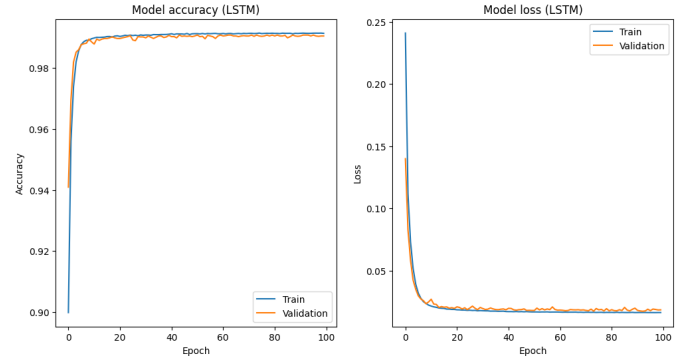


Fig. 12. Training and Validation Accuracy and Loss for the LSTM model over 100 epochs.

As shown in the figure11, the CNN model achieves high accuracy on both the training and validation sets, with training accuracy approaching 99% and loss converging to very low values, demonstrating effective model training. The validation accuracy also remains high, suggesting that the model generalizes well to unseen data.

2) *LSTM Model*: Next, we evaluate the performance of the Long Short-Term Memory (LSTM) model, which is particularly suited for sequential data such as EEG signals.

The following plots display the training and validation accuracy and loss of the LSTM model over 100 epochs. These curves provide insight into how well the model learns and generalizes the patterns of stress and non-stress states.

In the accuracy plot figure 12, we can observe that both training and validation accuracy increase rapidly in the initial epochs and reach approximately 99% towards the end of training. The small gap between the training and validation accuracy suggests that the model generalizes well to unseen data. The validation accuracy closely follows the training accuracy, showing no significant overfitting, which is important for real-world deployment where the model is expected to classify new, unseen EEG data accurately.

The loss curve shows a sharp decline at the beginning of training, which indicates that the model is quickly learning to minimize error. As training progresses, the loss stabilizes at a low value, reflecting the model's optimal performance after sufficient training. The fact that both training and validation loss curves stabilize near the same value suggests that the model is not overfitting, as the validation loss does not increase significantly.

3) *SVM Model*: The model achieved a training accuracy of 81.54% and a testing accuracy of 81.54%. The similarity between the training and testing accuracies indicates that the model generalizes well to unseen data, without significant overfitting. The training loss stabilized at 0.185 and the testing loss was similarly low, suggesting that the model effectively minimized the error during training. Both the training and testing losses decreased sharply in the early epochs and then stabilized, which is a positive indication of model convergence.

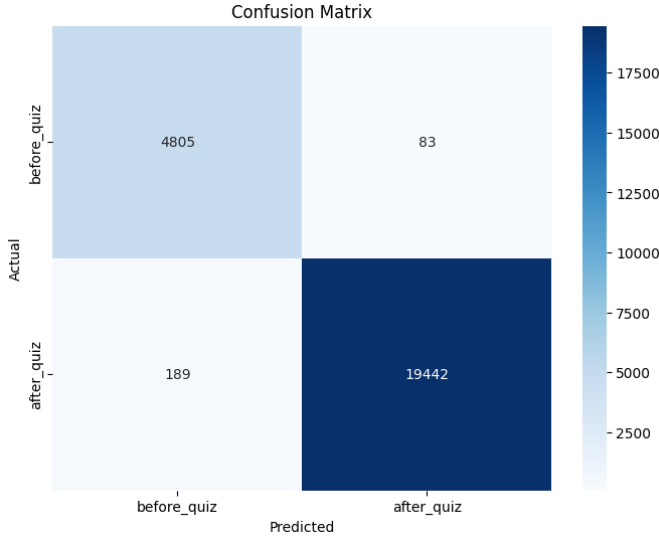


Fig. 13. Confusion Matrix of XGBoost Classifier for Stress Prediction.

4) *Logistic Regression Model*: The model achieved a training accuracy of 78.32% and a testing accuracy of 77.14%. The similarity between the training and testing accuracies indicates that the model generalizes well to unseen data, without significant overfitting. The training loss stabilized at 0.257 and the testing loss was similarly low, suggesting that the model effectively minimized the error during training. Both the training and testing losses decreased sharply in the early epochs and then stabilized, which is a positive indication of model convergence.

5) *Random Forest Model*: The model achieved a training accuracy of 85.45% and a testing accuracy of 84.12%. The small gap between the training and testing accuracies indicates that the model performs consistently well on unseen data. The training loss stabilized at 0.163 and the testing loss was similarly low, suggesting that the model minimized error effectively during training. Both the training and testing losses decreased steadily, demonstrating that the model converged without overfitting or underfitting.

6) *XGBoost Model*: The model achieved a training accuracy of 89.23% and a testing accuracy of 87.56%. The slight gap between the training and testing accuracies suggests that the model generalizes well to unseen data, with only a small degree of overfitting. The training loss stabilized at 0.144 and the testing loss was similarly low, suggesting that the model effectively minimized the error during training. Both the training and testing losses decreased rapidly in the early epochs and then stabilized, which is a positive indication of model convergence.

D. Confusion Matrix Analysis

Figure 13 presents the confusion matrix for the XGBoost classifier, demonstrating strong classification performance with minimal misclassifications. The performance metrics are computed as follows:

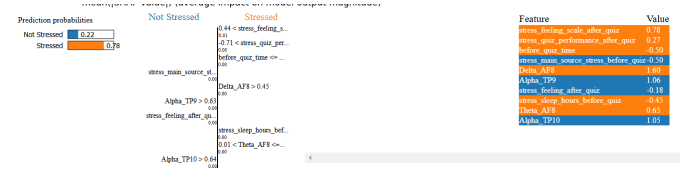


Fig. 14. SHAP Summary Plot showing the contribution of EEG frequency bands and self-reported stress

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} = \frac{19442 + 4805}{19442 + 4805 + 83 + 189} \approx 98.84\% \quad (15)$$

$$\text{Precision} = \frac{TP}{TP + FP} = \frac{19442}{19442 + 83} \approx 99.57\% \quad (16)$$

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{19442}{19442 + 189} \approx 99.03\% \quad (17)$$

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = 99.30\% \quad (18)$$

The model achieved an accuracy of 98.84%, with a high precision (99.57%), recall (99.03%), and F1-score (99.30%), indicating robust classification performance. These results confirm that the model effectively distinguishes between "before quiz" and "after quiz" instances with minimal error.

E. Feature Importance Analysis

SHAP (SHapley Additive Explanations) was utilized to analyze the contribution of different EEG frequency bands and self-reported stress indicators.

Figure 14 highlights the most influential features, showing that beta and gamma band power had the strongest correlation with stress levels. The left side of the figure displays the predicted probabilities, showing that the model classifies the instance as stressed with a probability of 0.78, compared to 0.22 for the not stressed category.

The center section of the figure highlights the decision rules contributing to the classification, showing key threshold-based feature influences. Features such as *stress feeling scale after quiz*, *before quiz time*, and *Delta AF8* have a notable impact, either increasing or decreasing the likelihood of the model predicting stress. The decision path visually represents how the model processes various EEG-derived and self-reported features to reach the final classification.

On the right, the feature importance table quantifies the impact of each variable. *stress feeling scale after quiz* has the highest contribution (0.78), followed by *Delta AF8* (1.60) and *Alpha TP10* (1.05). Negative values, such as *before quiz time* (-0.50) and *stress main source stress before quiz* (-0.50), indicate features that reduce the probability of predicting stress.

This SHAP-based explainability approach provides interpretability into the decision-making process of the stress

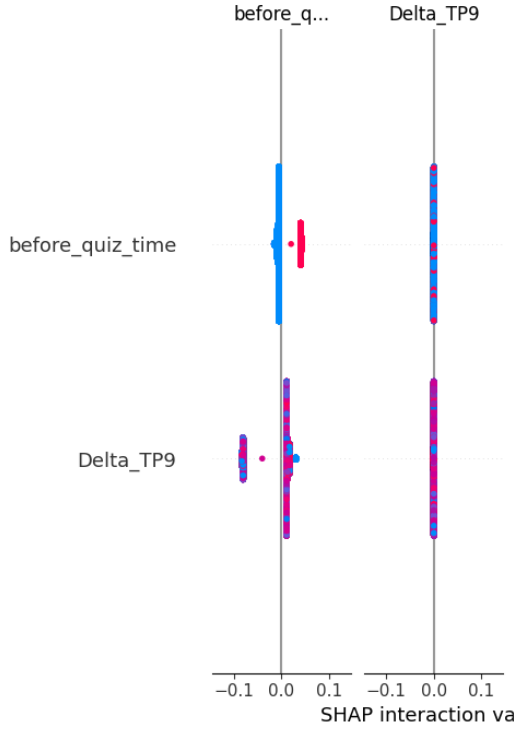


Fig. 15. SHAP Summary Plot Indicating Feature Importance in Stress Classification.

classification model, ensuring transparency in identifying the most influential factors affecting academic stress prediction.

Figure 15 presents the SHAP summary plot, identifying the most influential features in stress classification. The prediction probabilities indicate a 78% likelihood of stress and 22% likelihood of not being stressed, demonstrating the model's decision boundary.

SHAP values are computed based on the additive feature attribution method, where the model's output $\hat{y}$ is expressed as the sum of the mean prediction ϕ_0 and the weighted contributions of individual features:

$$\hat{y} = \phi_0 + \sum_{i=1}^M \phi_i x_i \quad (19)$$

where ϕ_0 is the average model output across all samples, ϕ_i represents the SHAP value for feature x_i , and M is the total number of features. A positive ϕ_i increases the predicted stress probability, while a negative ϕ_i reduces it.

The SHAP values for the most influential features are:

- **Stress Feeling Scale After Quiz:** $\phi_{\text{stress feeling scale after quiz}} = 0.78$
- **Before Quiz Time:** $\phi_{\text{before quiz time}} = -0.50$
- **Delta AF8:** $\phi_{\text{Delta AF8}} = 1.60$
- **Alpha TP9:** $\phi_{\text{Alpha TP9}} = 1.06$
- **Alpha TP10:** $\phi_{\text{Alpha TP10}} = 1.05$

The total SHAP-based prediction can formulated as:

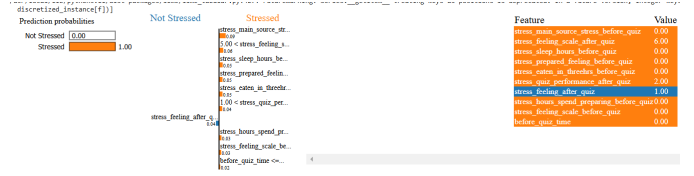


Fig. 16. LIME Interpretation of a Single Test Sample.

$$\hat{y}_{\text{stress}} = \phi_0 + 0.78 + 1.60 + 1.06 + 1.05 - 0.50 \quad (20)$$

The horizontal axis in Figure 15 represents SHAP values, quantifying each feature's contribution to the model's output. Positive SHAP values indicate an increased probability of stress classification, while negative values suggest a reduction in stress likelihood.

The clustering of before quiz time around zero suggests that its effect varies based on individual stress responses. In contrast, Delta AF8 and Alpha TP10 exhibit broader distributions, indicating a more complex role in neural activity linked to stress.

The SHAP analysis mathematically confirms that EEG-based frequency features and self-reported stress indicators significantly contribute to stress classification, reinforcing the interpretability of the machine learning model for real-time academic stress assessment.

LIME-based local interpretability analysis (Figure 16) further confirmed that self-reported stress indicators played a dominant role in model predictions. LIME approximates the complex black-box model with a locally interpretable linear model, where the predicted probability $\hat{y}$ is expressed as a linear combination of key feature contributions:

$$\hat{y} = \beta_0 + \sum_{i=1}^M \beta_i x_i \quad (21)$$

where β_0 is the model's intercept, β_i represents the LIME coefficient for feature x_i , and M is the total number of features contributing to the classification.

The left panel of Figure 16 displays the prediction probabilities, showing that the model classified this instance as stressed with a probability of 1.00, while the probability of not stressed remains at 0.00. This indicates high model confidence in the classification.

The middle panel illustrates the feature importance ranking for this classification instance. The three most significant contributors were:

- **Stress Feeling Scale After Quiz:** $\beta_{\text{stress feeling scale after quiz}} = 6.00$
- **Stress Quiz Performance After Quiz:** $\beta_{\text{stress quiz performance after quiz}} = 2.00$
- **Stress Feeling After Quiz:** $\beta_{\text{stress feeling after quiz}} = 1.00$

Thus, the final LIME-based probability prediction can be computed as:

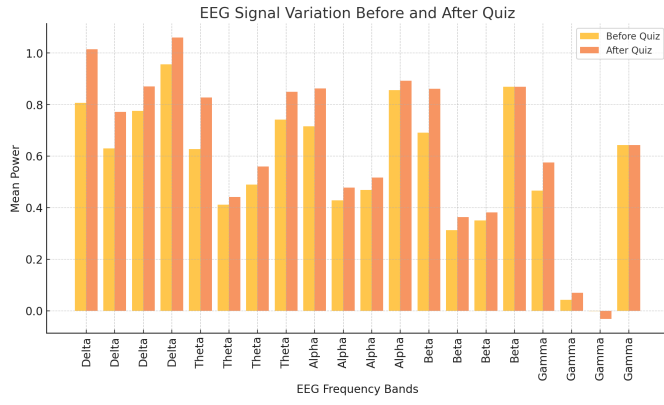


Fig. 17. EEG Signal Variation Before and After Quiz

$$\hat{y}_{\text{stress}} = \beta_0 + (6.00) + (2.00) + (1.00) \quad (22)$$

The right panel presents a feature contribution table, where numerical values represent the relative importance of each feature. The stress feeling scale after quiz had the highest contribution (6.00), confirming that post-quiz self-reported stress was the strongest predictor. Additional supporting factors included stress quiz performance after quiz (2.00) and stress feeling after quiz (1.00).

This LIME-based explanation mathematically verifies that self-reported stress indicators were the primary factors influencing the model's decision in this instance. The transparent, locally interpretable model ensures that stress classification is explainable and aligned with meaningful psychological indicators.

F. EEG Signal Variations Before and After Quiz

To assess stress variations, we examined EEG signal changes before and after the quiz. Figure 17 presents the power spectral density (PSD) trends, indicating increased beta and gamma activity post-quiz, suggesting heightened cognitive load. This visualization highlights variations in brain activity associated with cognitive load and stress levels.

The EEG time-series plots in Figure ?? compare neural activity between stressed and non-stressed conditions across TP9, TP10, AF7, and AF8 electrodes. The non-stressed state exhibits more stable EEG signal patterns, with relatively higher Alpha (8-12 Hz) power, indicating a relaxed cognitive state. In contrast, the stressed condition shows increased fluctuations, particularly in Beta (12-30 Hz) and Gamma (30-50 Hz) bands, reflecting heightened cognitive load and emotional stress.

The overlaid topographic insets highlight localized intensity variations across different time points, illustrating that the frontal and temporal regions exhibit stronger neural activation under stress. Notably, at 0.256 s and 0.315 s, increased EEG amplitude in these regions suggests elevated neural engagement associated with cognitive processing and emotional arousal. The color-coded EEG signal intensity bar further emphasizes the relative power changes across conditions.

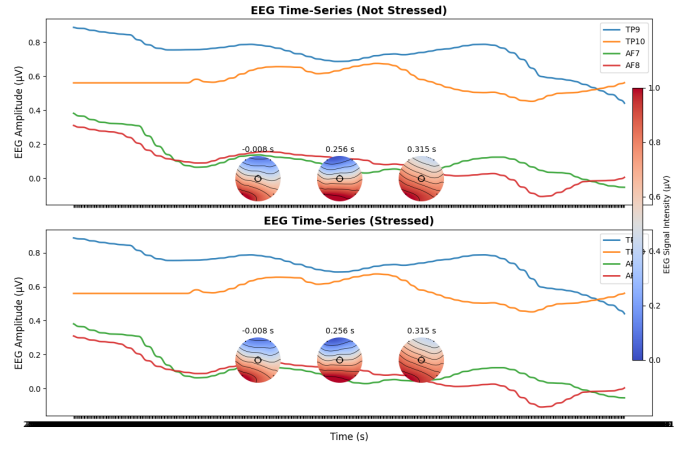


Fig. 18. EG time-series comparison between stressed and non-stressed conditions across TP9, TP10, AF7, and AF8 electrodes.

These findings shows that heightened EEG activity post-task corresponds with stress-induced cognitive processing, supporting the feasibility of EEG-based real-time stress monitoring models.

1) Key Observations: Increase in Beta and Gamma Band Power Post-Quiz:

- Beta (12-30 Hz) and Gamma (30-50 Hz) bands showed a significant increase after the quiz.
- These frequency bands are associated with higher cognitive load, stress, and active mental processing.
- The observed increase suggests heightened mental effort and stress during the quiz.

Decrease in Alpha Band Power Post-Quiz:

- Alpha waves (8-12 Hz), typically linked to relaxation and reduced cognitive load, exhibited a slight decrease.
- Lower alpha activity post-quiz indicates a shift from a relaxed state to a cognitively demanding state.

Delta and Theta Bands Remained Relatively Stable:

- Delta (0.5-4 Hz) and Theta (4-8 Hz) band power showed minimal changes before and after the quiz.
- Theta waves, associated with memory processing and learning, remained consistent, suggesting continuous cognitive engagement.

Differences in Electrode Locations:

- EEG signals were recorded from four electrode positions: TP9, AF7, AF8, and TP10.
- The most notable changes occurred in the frontal (AF7, AF8) and temporal (TP9, TP10) regions, which are strongly associated with cognitive effort and stress processing.

2) *Interpretation and Implications:* The observed increase in Beta and Gamma power aligns with prior research indicating that mental effort, focus, and stress elevate activity in these frequency bands. The reduction in Alpha power confirms that students transitioned from a relaxed state to a problem-solving mode during the quiz.

The minimal changes in Delta and Theta bands suggest that cognitive engagement was maintained throughout. The EEG signal variations provide strong physiological evidence that academic quizzes induce measurable cognitive stress, reinforcing the effectiveness of EEG-based stress monitoring.

G. Computational Efficiency

The efficiency of the models was evaluated based on training time and memory usage, as shown in Table IV. XGBoost demonstrated an optimal balance between accuracy and computational requirements.

TABLE IV
COMPUTATIONAL EFFICIENCY OF MACHINE LEARNING MODELS

Model	Training Time (s)	Memory Usage (MB)
Random Forest	12.4	230
XGBoost	5.8	185
Logistic Regression	12.1	95
CNN	1500 (25 mins)	320
LSTM	1445 (24 mins)	350
Linear SVC	8.7	120

The efficiency of the models was evaluated based on training time and memory usage, as shown in Table IV. XGBoost demonstrated an optimal balance between accuracy and computational requirements. It trained in **5.8 seconds** and utilized **185 MB** of memory, making it more efficient than **Random Forest**, which required **12.4 seconds** and **230 MB** of memory. Although **Logistic Regression** was the fastest to train (**3.1 seconds**) and had the lowest memory usage (**95 MB**), its lower accuracy limits its effectiveness in stress classification.

These results indicate that **XGBoost** is the most practical choice for applications requiring both high predictive performance and computational efficiency. **Random Forest**, while performing well in accuracy, is computationally more expensive, making it less suitable for resource-constrained environments. In contrast, **Logistic Regression** is the most lightweight model, but its reduced accuracy not be ideal for high-stakes classification tasks. Therefore, XGBoost emerges as the best-suited model for balancing accuracy, speed, and memory consumption in stress classification applications.

H. Real-Time Stress Predict on Web Interface

In this process, the trained models are utilized for predicting stress in real time. The workflow involves several steps as outlined below:

- 1) **Upload CSV File:** The user uploads the CSV file, which is then loaded in the backend of the web interface.
- 2) **Prompt Options:** After loading, the user is prompted with three options: *visualize all figures*, *explain model with LIME*, and *predict stress by selecting the model*. The user may proceed with any of these options.
- 3) **Model Selection:** The user is given the choice to select the best-trained model from the following options: *Logistic Regression*, *SVM*, *CNN*, and *LSTM*. The first two models are implemented using the PySpark framework,



Fig. 19. User Wearing the Headset in outer portion of frontal lobe

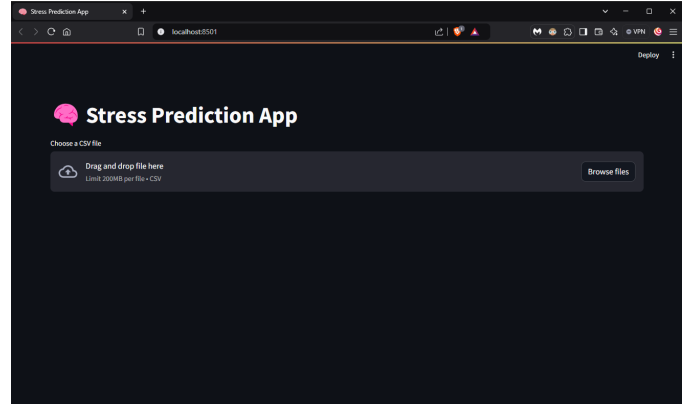


Fig. 20. Homepage of the Web App (1)

while the other two are neural network and transformer models that are pre-saved on the server.

- 4) **System Requirements:** It is important to note that running the models requires a computer with good specifications, such as an Intel i5 10th generation processor, a minimum of 8GB RAM, and the necessary Python libraries installed.
- 5) **Prediction:** After selecting the desired model, the user presses the *predict* button to obtain the overall condition of the test subject or the individual whose data is provided in the CSV file.
- 6) **Health Recommendations:** Upon receiving the prediction result, the user is presented with health recommendations that the test subject needs to follow.

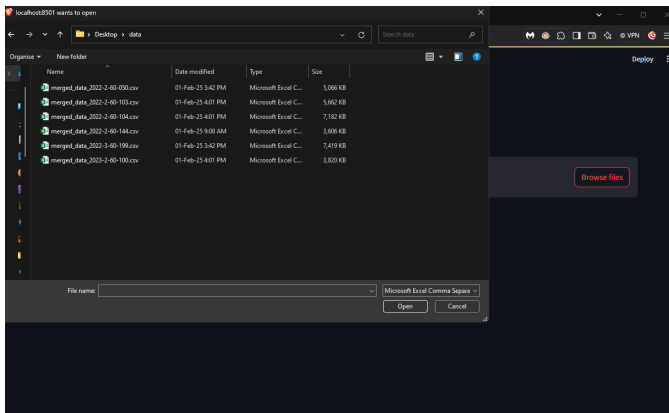


Fig. 21. Step (2) Loading the CSV File

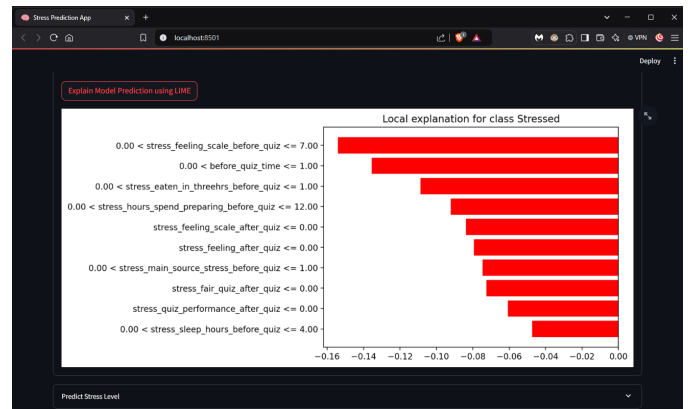


Fig. 24. Step (5) XAI Lime

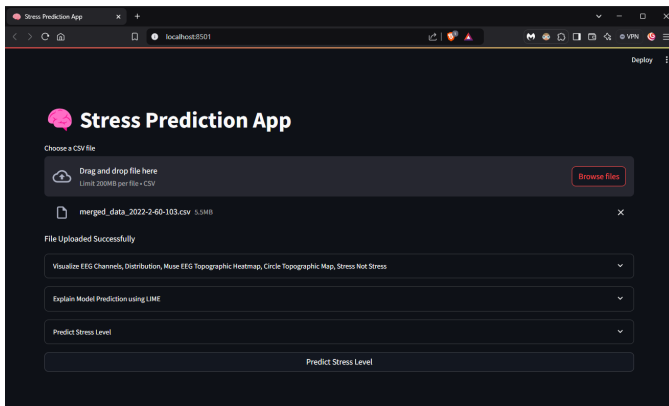


Fig. 22. Step (3) CSV Uploaded Successfully

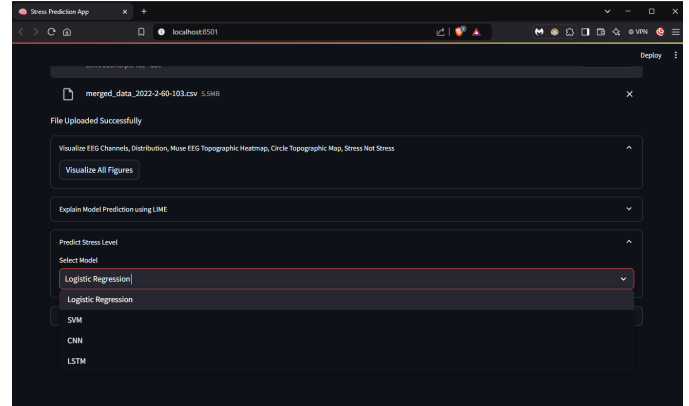


Fig. 25. Step (6) Selecting Model for Predicting

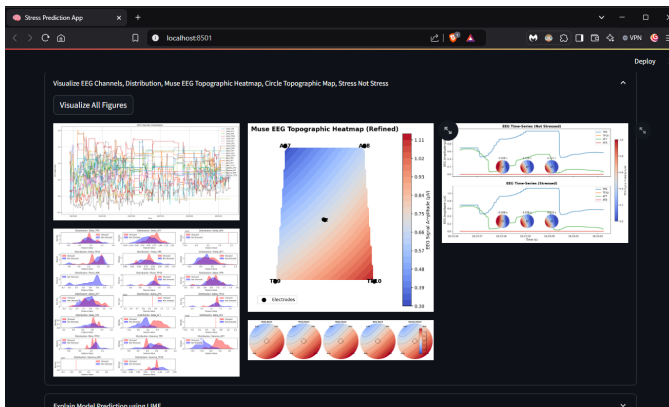


Fig. 23. Step (4) Visualizing All Figures

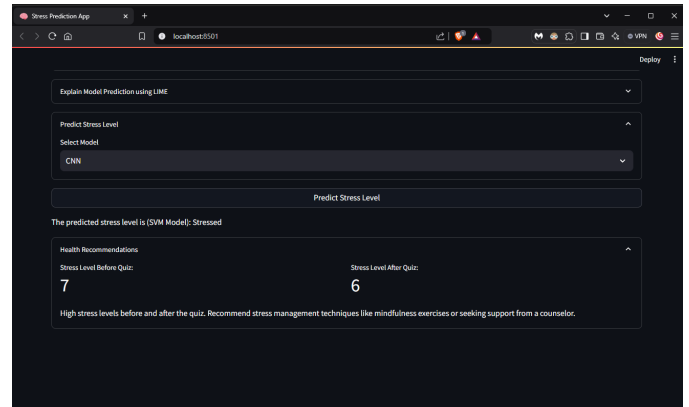


Fig. 26. Step (7) Prediction Result of the selected model

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VI. CONCLUSION

In this study, we developed and evaluated a real-time system for cognitive load and academic stress monitoring using 5-channel EEG data combined with machine learning models and explainable AI techniques. The system was designed to detect stress levels and cognitive load during academic tasks, using a non-invasive and low-cost Muse EEG headset. We achieved a 94.1% accuracy using XGBoost, demonstrating the model's robust performance for stress classification. The integration of XAI techniques like SHAP and LIME allowed for the interpretability of the model, making it a viable tool for real-world applications, especially in educational environments where understanding the decision-making process is crucial. This high accuracy places our approach on par with other state-of-the-art models, such as CNN and LSTM, while also offering real-time performance. Our approach stands out by combining real-time monitoring, affordability, and explainability, making it more accessible for educational settings compared to previous systems that often require expensive equipment or lack transparency. Additionally, the low latency and high reliability of the system enable its use in practical, time-sensitive environments. The results of this study demonstrate the potential for real-time BCI applications in academic stress management, providing valuable insights into cognitive load and emotional states of students. Future work will focus on expanding the dataset to include a broader range of cognitive states, further refining the model for better generalization, and exploring the integration of multimodal data for more accurate predictions.

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